

Note 5.3 The Definite Integral

Definition of the Definite Integral

- **Definition:**

Let $f(x)$ be a function defined on a closed interval $[a, b]$ and J is the limit of the Riemann sums $\sum_{k=1}^n f(c_k) \Delta x_k$, we say that J is the **Definite integral** of f on the interval $[a, b]$ if the following condition is satisfied:

given any number $\epsilon > 0$ there is a corresponding number $\delta > 0$ such that for every partition $P = \{x_0, x_1, \dots, x_n\}$ of $[a, b]$ with $\|P\| < \delta$ and any choice c_k in $[x_{k-1}, x_k]$, we have

$$\left| \sum_{k=1}^n f(c_k) \Delta x_k - J \right| < \epsilon$$

- **Understanding:**

The condition is a limiting process, in which the norm $\|P\|$ responds to the independent variable x , $x \rightarrow a$ responds to $\|P\| \rightarrow 0$; the Riemann sums $\sum_{k=1}^n f(c_k) \Delta x_k$ responds to the function $f(x)$ since the Riemann sums depends on the norm $\|P\|$; the J responds to the L .

And in the limiting process, it shows the specialization of the definite integral that the partition is arbitrary but it will still obtain the same limit J .

- **Remarks:**

In the cases where the subintervals all have equal width $\Delta x = (b - a)/n$, the Riemann sum can be represented as

$$S_n = \sum_{k=1}^n f(c_k) \Delta x_k = \sum_{k=1}^n f(c_k) \Delta x = \sum_{k=1}^n f(c_k) \left(\frac{b-a}{n} \right)$$

then the **Definite Integral** of f over $[a, b]$ is J that

$$J = \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n f(c_k) \Delta x_k = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(c_k) \Delta x$$

Note that it is not appropriate to divide the subintervals with equal width. Such as $f(x) = \frac{1}{x}$, $x \in [1, 2]$. We can divide the subinterval by proportional segmentation. ($2^{1/n}, \dots, 2^{k/n}$), then the Lower Sum is

$$\begin{aligned} \sum_{k=1}^n \frac{1}{2^{k/n}} \cdot (2^{k/n} - 2^{(k-1)/n}) &= \sum_{k=1}^n 1 - 2^{-\frac{1}{n}} \\ &= n \left(1 - 2^{-\frac{1}{n}} \right) \end{aligned}$$

Then since $||P|| = 2 - 2^{\frac{n}{n-1}}$, thus when $||P|| \rightarrow 0$, $n \rightarrow +\infty$. Then

$$J = \lim_{n \rightarrow +\infty} n \left(1 - 2^{-\frac{1}{n}} \right) = \lim_{t \rightarrow 0^+} \frac{1 - 2^{-t}}{t} = \ln 2$$

- **Notation:**

In the definition of the definite integral, the symbol for the number J is

$$\int_a^b f(x)dx$$

in which the integral symbol $\int$ replaces the sum symbol $\sum$ in the limit, the continuous selection of the function value $f(x)$ replaces the function value $f(c_k)$ and the differential dx replace the subinterval widths Δx_k . The upper limit of integration b and the lower limit of integration a replaced the interval $[a, b]$.

And we say the Riemann sums of f over $[a, b]$ **converge** to the definite integral

$J = \int_a^b f(x)dx$ and f is **integrable** over $[a, b]$. The variable of integration is called **dummy variable** representing the real number in the closed interval $[a, b]$.

Integrable and Nonintegrable Functions

- **Theorem 1: Integrability of continuous functions**

If a function f is continuous over $[a, b]$ or if f only has finitely discontinuities there, then the definite integral $\int_a^b f(x)dx$ exists and f is integrable over $[a, b]$.

- **Nonintegrable Functions:**

A function that is sufficiently discontinuous such that the area of the graph cannot be approximated by rectangles is nonintegrable function.

Properties of Definite Integrals

- **Rules satisfied by definite integrals:**

When f and g are integrable over $[a, b]$, the definite integral satisfies the underlying rules:

- Order of integration:

$$\int_b^a f(x)dx = - \int_a^b f(x)dx$$

(Definition as above is convenient to adapting to the algorithm laws)

- Zero width interval:

$$\int_a^a f(x)dx = 0$$

- Constant Multiple:

$$\int_a^b k f(x) dx = k \int_a^b f(x) dx$$

- Sum and difference:

$$\int_a^b (f(x) \pm g(x)) dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$$

- Additivity:

$$\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx$$

- Max-Min inequality:

If f has maximum value $\max f$ and the minimum value $\min f$ on $[a, b]$, then

$$\min f \leq \frac{\int_a^b f(x) dx}{b - a} \leq \max f$$

that is,

$$\min f \cdot (b - a) \leq \int_a^b f(x) dx \leq \max f \cdot (b - a)$$

- Domination:

$$f(x) \geq g(x) \text{ on } [a, b] \implies \int_a^b f(x) dx \geq \int_a^b g(x) dx$$

$$f(x) \geq 0 \text{ on } [a, b] \implies \int_a^b f(x) dx \geq 0$$

idea: When the problem is difficult, we can first find a special case that is easy. Then we can use the special case to prove the problem (Rolle's Theorem and Mean Value Theorem)

Area Under the Graph of a Nonnegative Function

- **Definition:**

If $y = f(x)$ is nonnegative and integrable on $[a, b]$, then the **area under the curve** $= ()$ on $[,]$ is the integral of f from a to b ,

$$A = \int_a^b f(x) dx$$

- Conclusion:

$$\int_a^b x \, dx = \frac{b^2}{2} - \frac{a^2}{2}, \quad a < b$$

$$\int_a^b c \, dx = c(b - a), \quad c \text{ any constant}$$

$$\int_a^b x^2 \, dx = \frac{b^3}{3} - \frac{a^3}{3}, \quad a < b$$

- Proof of the third conclusion:

First we define $f(x) = x^2$ on $[0, b]$. Then we subdivided the interval into n equal subintervals and $\Delta x = \frac{b}{n}$. And we choose c_k as the right endpoint in each subinterval so the partition is $P = \{0, \frac{b}{n}, \frac{2b}{n}, \dots, \frac{nb}{n}\}$ and $c_k = \frac{kb}{n}$. So

$$\begin{aligned} \sum_{k=1}^n f(c_k) \Delta x &= \sum_{k=1}^n \left(\frac{kb}{n} \right)^2 \cdot \frac{b}{n} \\ &= \frac{b^3}{n^3} \sum_{k=1}^n k^2 \\ &= \frac{b^3}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} \\ &= \frac{b^3}{6} \cdot \left(2 + \frac{3}{n} + \frac{1}{n^2} \right) \end{aligned}$$

As $n \rightarrow \infty$, the last expression has the limit $b^3/3$. Therefore,

$$\int_0^b x^2 \, dx = \frac{b^3}{3}$$

Thus

$$\begin{aligned} \int_a^b x^2 \, dx &= \int_a^0 x^2 \, dx + \int_0^b x^2 \, dx \\ &= - \int_0^a x^2 \, dx + \int_0^b x^2 \, dx \\ &= \frac{b^3}{3} - \frac{a^3}{3} \end{aligned}$$

Average Value of a Continuous Function Revisited

- Two approaches to obtain the average:
 - Define the average as the area under the graph of $f(x)$ on $[a, b]$ divided by $b - a$.
 - Define the average as n numbers' sum divided by n :

We subdivide $[a, b]$ into n subintervals of equal width $\Delta x = (b - a)/n$. Then the

average of the n sampled values is

$$\begin{aligned}\frac{f(c_1) + f(c_2) + \cdots + f(c_n)}{n} &= \frac{1}{n} \sum_{k=1}^n f(c_k) \\ &= \frac{\Delta x}{b-a} \sum_{k=1}^n f(c_k) \\ &= \frac{1}{b-a} \sum_{k=1}^n f(c_k) \Delta x\end{aligned}$$

As the size of sample increases, the norm of the partition approach zero such that the average approaches the definition.

- **Definition:**

If f is integrable on $[a, b]$, then its average value on $[a, b]$, also called its **mean**, is

$$\text{av}(f) = \frac{1}{b-a} \int_a^b f(x) dx$$